

Basis and Dimension II

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1 Dimensions of standard spaces

Recall that for a vector space V over F ,

- S is a basis for V if S is linearly independent such that $V = \text{span}(S)$;
- $\dim_F(V) = |S|$.

(Sometimes we may suppress $\dim_F$ as just $\dim$ if the underlying field is clear.)

1.1 Example

1. (Vectors)

$$V = F^n$$

is a vector space over F .

$$\dim_F(F^n) = n.$$

The standard basis is

$$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\},$$

where

$$\{\mathbf{e}_1 = (1, 0, \dots, 0), \mathbf{e}_2 = (0, 1, \dots, 0), \dots, \mathbf{e}_n = (0, 0, \dots, 1)\}.$$

2. (Matrix spaces)

$$\dim_F(M_n(F)) = n^2.$$

The “standard” basis is

$$\{E_{11}, E_{12}, \dots, E_{nn}\},$$

where E_{ij} is the matrix whose (i, j) entry is 1 and all other entries are 0.
i.e.

$$E_{11} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \quad E_{12} = \begin{pmatrix} 0 & 1 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \quad \dots \quad E_{nn} = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

3.

Find a basis and the dimension of the following subspace of $\mathbb{R}^4$:

$$W = \{(x, y, z, w) \in \mathbb{R}^4 : x + y + z + w = 0 \text{ and } -x + 2z + 3w = 0\}.$$

Step 1: Simplifying the equations

$$\begin{cases} x + y + z + w = 0 & (1) \\ -x + 2z + 3w = 0 & (2) \end{cases}$$

Adding (1) and (2), we obtain

$$\begin{cases} x + y + z + w = 0 & (1) \\ y + 3z + 4w = 0 & (2)' \end{cases}$$

Step 2: Backward substitution

By (2)',

$$y + 3z + 4w = 0.$$

Let

$$z = s, \quad w = t.$$

Then

$$y = -3s - 4t.$$

By (1),

$$x + y + z + w = 0,$$

so

$$\begin{aligned} x + (-3s - 4t) + s + t &= 0 \\ x - 2s - 3t &= 0 \\ x &= 2s + 3t. \end{aligned}$$

Therefore

$$W = \{(2s + 3t, -3s - 4t, s, t) : s, t \in \mathbb{R}\}.$$

$$= \{s(2, -3, 1, 0) + t(3, -4, 0, 1) : s, t \in \mathbb{R}\}.$$

$$= \text{span}\{(2, -3, 1, 0), (3, -4, 0, 1)\}.$$

As

$$B = \{(2, -3, 1, 0), (3, -4, 0, 1)\}$$

spans W , and B is linearly independent, by def, B forms a basis of W .

Therefore,

$$\dim_{\mathbb{R}}(W) = 2.$$

4. (Polynomial of degree at most n)

$$\dim_F(F[x]_{\leq n}) = n + 1.$$

The “standard” basis is

$$\{1, x, x^2, \dots, x^n\}.$$

5. (Zero space)

Recall that

$$\{0\} = \text{span}(\emptyset),$$

where

$$\emptyset$$

denotes the empty set.

Therefore,

$$\emptyset$$

forms a basis of

$$\{0\}.$$

Hence,

$$\dim_F(\{0\}) = |\emptyset| = 0.$$

6. (Complex numbers)

$$\mathbb{C} = \{a \cdot 1 + b \cdot \sqrt{-1} : a, b \in \mathbb{R}\}.$$

Hence,

$$\dim_{\mathbb{R}}(\mathbb{C}) = 2.$$

On the other hand,

$$\mathbb{C} = \{a \cdot 1 : a \in \mathbb{C}\}.$$

Hence,

$$\dim_{\mathbb{C}}(\mathbb{C}) = 1.$$

2 Consequence of Replacement Theorem

2.1 Theorem

Theorem 3.2.1.

Let V be a vector space over F of dimension n , and let $S \subseteq V$.

1. If $|S| > n$, then S cannot be linearly independent.
2. If $|S| < n$, then S cannot span V .

(1)

Proof. Given $|S| > n$.

Suppose, for contradiction, that S is linearly independent.

Let B be a basis of V . Then

$$|B| = n.$$

Since B spans V , the Replacement Theorem implies

$$|S| \leq |B| = n,$$

which contradicts $|S| > n$.

Therefore, S cannot be linearly independent.

□

(2)

Proof. Given $|S| < n$.

Suppose, for contradiction, that S spans V .

Let B be a basis of V . Then

$$|B| = n.$$

Since B is linearly independent and S spans V , the Replacement Theorem implies

$$|B| \leq |S|.$$

Hence

$$n = |B| \leq |S|,$$

which contradicts $|S| < n$.

Therefore, S cannot span V

□

Theorem 3.2.2. Let V and W be finite-dimensional vector spaces.
If

1. $W \subseteq V$,
2. $\dim_F(W) = \dim_F(V)$,

then

$$W = V.$$

Proof. Given

$$W \subseteq V$$

and

$$\dim_F(W) = \dim_F(V) = n.$$

We WTS that $V \subseteq W$.

Suppose not. Then there exists

$$\mathbf{v} \in V$$

such that

$$\mathbf{v} \notin W.$$

Let

$$S = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$$

be a basis of W .

Since

$$W = \text{span}(S),$$

we have

$$\mathbf{v} \notin \text{span}(S).$$

Therefore,

$$S' = \{\mathbf{w}_1, \dots, \mathbf{w}_n, \mathbf{v}\}$$

is linearly independent in V .

Since

$$\dim_F(V) = n,$$

Theorem 3.2.1 implies that any linearly independent subset of V contains at most n vectors.

However,

$$|S'| = n + 1,$$

a contradiction.

Hence

$$V \subseteq W.$$

Together with

$$W \subseteq V,$$

we conclude that

$$W = V.$$

□

Theorem 3.2.3. Let V be a vector space of dimension n . Then any n linearly independent vectors form a basis of V .

(In literature, we say that any maximal linearly independent subset forms a basis.)

Proof. Let S be a linearly independent subset of V with

$$|S| = n.$$

We WTS that

$$\text{span}(S) = V.$$

Let

$$W = \text{span}(S).$$

Then W is a subspace of V , so

$$W \subseteq V.$$

Since S spans W and is linearly independent, S is a basis of W .

Therefore,

$$\dim_F(W) = |S| = n.$$

Since

$$\dim_F(V) = n,$$

we have

$$\dim_F(W) = \dim_F(V).$$

By Theorem 3.2.2,

$$W = V.$$

Hence

$$\text{span}(S) = V.$$

Therefore S is a basis of V .

□

2.2 Example

1. $S_1 = \{(1, 0, 1), (1, 1, 0), (2, 3, 1), (-1, 2, 3)\}$ cannot be linearly independent.

$$S_1 = \{(1, 0, 1), (1, 1, 0), (2, 3, 1), (-1, 2, 3)\}.$$

Since

$$\dim_{\mathbb{R}}(\mathbb{R}^3) = 3,$$

Theorem 3.2.1 implies that any linearly independent subset of $\mathbb{R}^3$ contains at most 3 vectors.

However,

$$|S_1| = 4.$$

Therefore S_1 cannot be linearly independent.

2. $S_2 = \{(1, 0, 1), (1, 1, 0)\}$ cannot span $\mathbb{R}^3$.

$$S_2 = \{(1, 0, 1), (1, 1, 0)\}.$$

Since

$$\dim_{\mathbb{R}}(\mathbb{R}^3) = 3,$$

Theorem 3.2.1 implies that any spanning subset of $\mathbb{R}^3$ must contain at least 3 vectors.

However,

$$|S_2| = 2.$$

Therefore S_2 cannot span $\mathbb{R}^3$.

3. The subset $S = \{(1, 0, 1), (1, 1, 0), (2, 3, 1)\}$ forms a basis of $\mathbb{R}^3$.

(Check.) S is linearly independent.

Since

$$\dim_{\mathbb{R}}(\mathbb{R}^3) = 3 = |S|,$$

S is a **maximal linearly independent set**. Therefore S is a basis of $\mathbb{R}^3$.

Remark: This avoids the tedious check of spanning.

4. Let $V = M_2(\mathbb{R})$ and consider the subset

$$S = \left\{ A_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, A_2 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, A_3 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, A_4 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, A_5 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, A_6 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \right\}.$$

- (a) Prove that $\text{span}(S) = V$

Proof. Let

$$S' = \{A_1, A_2, A_3, A_4\}.$$

Suppose

$$a_1A_1 + a_2A_2 + a_3A_3 + a_4A_4 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

By comparing the entries, we obtain

$$\begin{cases} a_1 + a_2 + a_3 = 0, \\ a_1 + a_4 = 0, \\ a_2 = 0, \\ a_3 + a_4 = 0. \end{cases}$$

Hence

$$a_1 = a_2 = a_3 = a_4 = 0.$$

Therefore S' is linearly independent.

Since

$$|S'| = 4 = \dim_{\mathbb{R}}(M_2(\mathbb{R})),$$

Theorem 3.2.3 implies that S' is a basis of V .

Thus

$$V = \text{span}(S').$$

Since

$$S' \subseteq S,$$

we conclude that

$$\text{span}(S) = V.$$

□

(b) Find a subset of S that is a basis of V

By part (a),

$$\{A_1, A_2, A_3, A_4\}$$

is a basis of V .

(c) Find a subset of S with 4 elements which is not a basis of V

Observe that

$$A_1 + A_6 = A_2 + A_4 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Therefore

$$A_1 - A_2 - A_4 + A_6 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Since the coefficients are not all zero,

$$\{A_1, A_2, A_4, A_6\}$$

is linearly dependent.

Hence

$$\{A_1, A_2, A_4, A_6\}$$

is not a basis of V .

3 Sifting and proof of Replacement Theorem

3.1 Theorem

Theorem 3.3.1. A subset

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$$

of V is linearly independent if and only if for each $r = 1, \dots, m$, $\mathbf{v}_r$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$. In particular, this requires $\mathbf{v}_1 \neq \mathbf{0}$.

Proof. This is just Theorem 2.2.1. □

Theorem 3.3.2. Let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m, \mathbf{w}\}.$$

If $\mathbf{w}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$, then

$$\text{span}(S) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}.$$

Proof. ($\subseteq$)

Given

$$\mathbf{w} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_m\mathbf{v}_m$$

for some $a_i \in F$.

We first show

$$\text{span}(S) \subseteq \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}.$$

Let $\mathbf{v} \in \text{span}(S)$. Then

$$\begin{aligned} \mathbf{v} &= b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \dots + b_m\mathbf{v}_m + c\mathbf{w} \\ &= b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \dots + b_m\mathbf{v}_m + c(a_1\mathbf{v}_1 + \dots + a_m\mathbf{v}_m) \\ &= (b_1 + ca_1)\mathbf{v}_1 + (b_2 + ca_2)\mathbf{v}_2 + \dots + (b_m + ca_m)\mathbf{v}_m. \end{aligned}$$

Hence

$$\mathbf{v} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\},$$

so

$$\text{span}(S) \subseteq \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}.$$

($\supseteq$)

Conversely, let

$$\mathbf{v} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}.$$

Then

$$\begin{aligned}\mathbf{v} &= d_1\mathbf{v}_1 + d_2\mathbf{v}_2 + \cdots + d_m\mathbf{v}_m \\ &= \underbrace{d_1\mathbf{v}_1 + d_2\mathbf{v}_2 + \cdots + d_m\mathbf{v}_m + 0 \cdot \mathbf{w}}_{\text{linear combination of } S}.\end{aligned}$$

Hence

$$\mathbf{v} \in \text{span}(S),$$

which implies

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\} \subseteq \text{span}(S).$$

Therefore,

$$\text{span}(S) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}.$$

□

Theorem 3.3.3. (Sifting method) Let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$$

be a subset of V . Then there exists a subset

$$S' \subseteq S$$

such that

1. S' is linearly independent; and
2. $\text{span}(S') = \text{span}(S)$.

Proof. For each r , check whether $\mathbf{v}_r$ is a linear combination of the previous vectors

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}.$$

1. Keep $\mathbf{v}_r$ in S if $\mathbf{v}_r$ is *not* a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$ (not redundant).
2. Remove $\mathbf{v}_r$ from S if $\mathbf{v}_r$ *is* a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$ (redundant).

After this process, we obtain a subset S' in which no element is a linear combination of the preceding members. Therefore, by Theorem 3.3.1, S' is linearly independent. Moreover, by Theorem 3.3.2, removing redundant members from S does not change its linear span. Hence, we have $\text{span}(S') = \text{span}(S)$.

□

Theorem 3.3.4. (Basis extension theorem)

Let V be a finite-dimensional vector space, and let S be a linearly independent subset of V . Then S can be extended into a basis of V .

Proof. Let

$$S = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$$

be a linearly independent subset of V .

Let

$$B = \{b_1, \dots, b_n\}$$

be any basis of V .

Consider

$$S \cup B = \{\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{b}_1, \dots, \mathbf{b}_n\}.$$

Apply the Sifting Method to $S \cup B$.

Since S is linearly independent, none of the vectors $\mathbf{v}_1, \dots, \mathbf{v}_m$ will be removed during the sifting process.

Hence we obtain

$$S \cup B', \quad \text{where } B' \subseteq B.$$

Notice that

1. $S \cup B'$ is linearly independent;
2. $\text{span}(S \cup B') = \text{span}(S \cup B)$.

Since B is a basis of V ,

$$\text{span}(S \cup B) = V.$$

Therefore,

$$\text{span}(S \cup B') = V.$$

Hence $S \cup B'$ is both linearly independent and spans V . Therefore,

$$S \cup B'$$

forms a basis of V , which is an extension of S . □

3.2 Proof of Exchange/Replacement Theorem

This is the proof of Replacement Theorem (Theorem 2.5.1) which says that if

- $S_1 = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a spanning set of V .
- $S_2 = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ is linearly independent.

Then

$$m \leq n.$$

Proof. Idea: Insert $\mathbf{w}_j$ one-by-one into S_1 , then apply the Sifting Method.

Consider

$$B_1 = \{\mathbf{w}_1, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \{\mathbf{w}_1\} \cup S_1.$$

Applying the Sifting Method to B_1 , we must remove some $\mathbf{v}_i$'s. Hence we obtain

$$B'_1 = \{\mathbf{w}_1\} \cup S'_1, \text{ where } |S'_1| < |S_1|.$$

Moreover,

$$B'_1$$

remains a spanning set of V .

Now consider

$$B_2 = \{\mathbf{w}_2, \mathbf{w}_1\} \cup S'_1.$$

Applying the Sifting Method again, none of the $\mathbf{w}_i$'s will be removed because

$$S_2 = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$$

is linearly independent.

Hence we obtain

$$B'_2 = \{\mathbf{w}_2, \mathbf{w}_1\} \cup S''_1, \text{ where } |S''_1| < |S'_1|.$$

Continue this process. Finally we obtain

$$B'_m = \{\mathbf{w}_m, \mathbf{w}_{m-1}, \dots, \mathbf{w}_1\} \cup S_1^{(m)},$$

where

$$0 \leq |S_1^{(m)}| < |S_1^{(m-1)}| < \dots < |S''_1| < |S'_1| < |S_1| = n.$$

Thus there are at least m distinct integers strictly less than n .

Therefore, we must have

$$m \leq n.$$

□

4 Direct sum of subspaces

4.1 Definition

Definition 3.4.1. (Sum of subspaces)

Let W_1, W_2 be subspaces of a vector space V . Then define

$$W_1 + W_2 := \{\mathbf{w}_1 + \mathbf{w}_2 : \mathbf{w}_1 \in W_1 \text{ and } \mathbf{w}_2 \in W_2\}$$

Definition 3.4.2. (Direct Sum)

We write $W_1 + W_2 = W_1 \oplus W_2$ if $W_1 \cap W_2 = \{\mathbf{0}\}$. In this case, we call it the *direct sum* (instead of just sum) of W_1 and W_2 .

4.2 Theorem

Theorem 3.4.1.

Let $W_1 + W_2$ be a subspaces of V . Indeed, this is smallest subspace containing both W_1 and W_2

Proof. We verify the three subspace conditions.

Non-empty

Since W_1 and W_2 are subspaces,

$$\mathbf{0} \in W_1, \quad \mathbf{0} \in W_2.$$

Hence

$$\mathbf{0} = \mathbf{0} + \mathbf{0} \in W_1 + W_2. \implies W_1 + W_2 \text{ is non-empty.}$$

Closed under addition

Let

$$\mathbf{a}, \mathbf{b} \in W_1 + W_2.$$

Then

$$\begin{cases} \mathbf{a} = \mathbf{w}_1 + \mathbf{w}_2 & \text{for some } \mathbf{w}_1 \in W_1 \text{ and } \mathbf{w}_2 \in W_2. \\ \mathbf{b} = \mathbf{w}'_1 + \mathbf{w}'_2 & \text{for some } \mathbf{w}'_1 \in W_1 \text{ and } \mathbf{w}'_2 \in W_2. \end{cases}$$

Therefore,

$$\mathbf{a} + \mathbf{b} = (\mathbf{w}_1 + \mathbf{w}'_1) + (\mathbf{w}_2 + \mathbf{w}'_2).$$

Since W_1 and W_2 are subspaces,

$$\mathbf{w}_1 + \mathbf{w}'_1 \in W_1, \quad \mathbf{w}_2 + \mathbf{w}'_2 \in W_2.$$

Hence

$$\mathbf{a} + \mathbf{b} \in W_1 + W_2. \implies W_1 + W_2 \text{ is Closed under addition}$$

Closed under scalar multiplication

Let

$$\mathbf{a} \in W_1 + W_2, \text{ and } c \text{ is a scalar}$$

Then

$$\begin{cases} \mathbf{a} = \mathbf{w}_1 + \mathbf{w}_2 \text{ for some } \mathbf{w}_1 \in W_1 \text{ and } \mathbf{w}_2 \in W_2 \\ c\mathbf{a} = c(\mathbf{w}_1 + \mathbf{w}_2) = c\mathbf{w}_1 + c\mathbf{w}_2 \text{ for some } \mathbf{w}_1 \in W_1 \text{ and } \mathbf{w}_2 \in W_2 \end{cases}$$

Since

$$c\mathbf{w}_1 \in W_1, \quad c\mathbf{w}_2 \in W_2,$$

we conclude that

$$c\mathbf{a} \in W_1 + W_2.$$

Therefore, $W_1 + W_2$ is a subspace of V . □

Theorem 3.4.2.

Let V be a finite-dimensional vector space, and let W_1, W_2 be subspaces of V . Then

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).$$

Proof. Idea. Skill for dimension formula: Start with the “smallest” subspace

$$W_1 \cap W_2 \subseteq W_1, \quad W_2 \subseteq W_1 + W_2.$$

Let

$$n = \dim(W_1 \cap W_2),$$

and let

$$B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$$

be a basis of $W_1 \cap W_2$.

Since $B \subseteq W_1$ and B is linearly independent, by the Basis Extension Theorem, we can extend B to a basis of W_1 :

$$B_1 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p\}, \text{ a basis of } W_1$$

Similarly, since $B \subseteq W_2$ and B is linearly independent, we may extend B to a basis of W_2 :

$$B_2 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{t}_1, \dots, \mathbf{t}_q\}, \text{ a basis of } W_2$$

BIG CLAIM

Let

$$S = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p, \mathbf{t}_1, \dots, \mathbf{t}_q\}.$$

We claim that S forms a basis of $W_1 + W_2$.

Proof of the claim. We verify the three subspace conditions:

(1) S spans $W_1 + W_2$

Let

$$\mathbf{w} \in W_1 + W_2.$$

Then there $\exists \mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W_2$ s.t.

$$\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2 \in W_1 + W_2$$

Since B_1 is a basis of W_1 ,

$$\mathbf{w}_1 = \sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j.$$

And B_2 is a basis of W_2 ,

$$\mathbf{w}_2 = \sum_{i=1}^n \alpha'_i \mathbf{b}_i + \sum_{k=1}^q \gamma_k \mathbf{t}_k.$$

So

$$\begin{aligned} \mathbf{w} &= \mathbf{w}_1 + \mathbf{w}_2 \\ &= \sum_{i=1}^n (\alpha_i + \alpha'_i) \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j + \sum_{k=1}^q \gamma_k \mathbf{t}_k. \end{aligned}$$

Therefore,

$$\mathbf{w} \in \text{span}(S).$$

Hence S spans $W_1 + W_2$.

(2) S is linearly independent.

Suppose $\exists$ scalars $\alpha_i, \beta_j, \gamma_k$ s.t.

$$\sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j + \sum_{k=1}^q \gamma_k \mathbf{t}_k = \mathbf{0}.$$

Rearranging,

$$\sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j = - \sum_{k=1}^q \gamma_k \mathbf{t}_k.$$

Let $\mathbf{v}$ be left-hand side. Undoubtedly, $\mathbf{v}$ equals right-hand side.

Clearly, $\mathbf{v}$ belongs to both W_1 and W_2 . Hence both sides belong to $W_1 \cap W_2$.

As

$$B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$$

is a basis of $W_1 \cap W_2$, $\mathbf{v}$ can be written as

$$\mathbf{v} = \sum_{i=1}^n \sigma_i \mathbf{b}_i \implies \mathbf{0} = \sum_{i=1}^n \sigma_i \mathbf{b}_i + \sum_{k=1}^q \gamma_k \mathbf{t}_k$$

As $B_2 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{t}_1, \dots, \mathbf{t}_q\}$ is linearly independent, this forces $\sigma_i = 0 \forall i$, $\gamma_k = 0 \forall k$

Then the rearranging part above can be reduced to

$$\sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j = \mathbf{0}.$$

As $B_1 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p\}$ is linearly independent, this forces $\alpha_i = 0 \forall i$, $\beta_j = 0 \forall j$

So S is linearly independent. □

Hence S is a basis of $W_1 + W_2$,

$$\begin{aligned} \dim(W_1 + W_2) &= |S| \\ &= n + p + q \\ &= (n + p) + (n + q) - n \\ &= \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2). \end{aligned}$$

□

Theorem 3.4.3.

If W_1, W_2 are finite-dimensional subspaces of V with $W_1 \cap W_2 = \{\mathbf{0}\}$, then

$$\dim(W_1 \oplus W_2) = \dim(W_1) + \dim(W_2).$$

Proof. Since

$$W_1 \oplus W_2 = W_1 + W_2,$$

By Theorem 3.4.2, we have

$$\begin{aligned} \dim(W_1 \oplus W_2) &= \dim(W_1 + W_2) \\ &= \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2) \\ &= \dim(W_1) + \dim(W_2) - \mathbf{0} \\ &= \dim(W_1) + \dim(W_2). \end{aligned}$$

□

Theorem 3.4.4.

Let V be a finite-dimensional vector space and let W be a subspace of V . Then there exists another subspace W' of V such that $V = W \oplus W'$.

In this case, we call W' A *complementary subspace* of W in V .

Proof. Let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$$

be a basis of W .

By the Basis Extension Theorem, we may extend S to a basis of V , and we say that

$$S' = \{\underbrace{\mathbf{v}_1, \dots, \mathbf{v}_k}_{\text{span } W}, \mathbf{s}_1, \dots, \mathbf{s}_n\}.$$

Define

$$W' = \text{span}\{\mathbf{s}_1, \dots, \mathbf{s}_n\}.$$

Let $V = W \oplus W'$, we claim that

$$(1) V \subseteq W + W', \quad (2) W \cap W' = \{\mathbf{0}\}.$$

proof of (1). Let

$$\mathbf{v} \in V.$$

As S' is a basis of V ,

$$\mathbf{v} = \underbrace{\sum_{i=1}^k \alpha_i \mathbf{v}_i}_{\in W} + \underbrace{\sum_{j=1}^n \beta_j \mathbf{s}_j}_{\in W'} \in W + W'$$

□

proof of (2). Let

$$\mathbf{w} \in W \cap W'.$$

Since both $\mathbf{w} \in W$ and $\mathbf{w} \in W'$,

$$\mathbf{w} = \sum_{i=1}^k \alpha_i \mathbf{v}_i = \sum_{j=1}^n \beta_j \mathbf{s}_j.$$

Hence,

$$\sum_{i=1}^k \alpha_i \mathbf{v}_i - \sum_{j=1}^n \beta_j \mathbf{s}_j = \mathbf{0}.$$

Since S' is linearly independent,

$$\alpha_i = \beta_j = 0, \quad \forall i, j$$

Thus, $\mathbf{w} = \mathbf{0}$, $\implies W \cap W' = \{\mathbf{0}\}$.

□

□

4.3 Question

Question: Is true that, $\dim(W_1 + W_2 + W_3) = \dim(W_1) + \dim(W_2) + \dim(W_3) - \dim(W_1 \cap W_2) - \dim(W_2 \cap W_3) - \dim(W_3 \cap W_1) + \dim(W_1 \cap W_2 \cap W_3)$?

Answer: **False**.

4.4 Example

4.4.1 Direct sum of symmetric and skew-symmetric matrices

Consider the vector space $V = M_n(\mathbb{C})$ and two of its subspaces :

- $W_1 = \{A \in M_n(\mathbb{C}) \mid A^t = A\}$
- $W_2 = \{A \in M_n(\mathbb{C}) \mid A^t = -A\}$

Prove that $V = W_1 \oplus W_2$.

Proof. We need to show

- (1) $V = W_1 + W_2$
i.e. any square matrix is a sum of a symmetric and a skew-symmetric matrix
- (2) $W_1 \cap W_2 = \{\mathbf{0}\}$ (**zero matrix**)

proof of (1). Let

$$A \in M_n(\mathbb{C}).$$

Observe that

$$A = \frac{1}{2}(A + A^t) + \frac{1}{2}(A - A^t).$$

Let

$$B = \frac{1}{2}(A + A^t), \quad C = \frac{1}{2}(A - A^t).$$

Then

$$B^t = \frac{1}{2}(A + A^t)^t = \frac{1}{2}(A^t + A) = B,$$

so B is symmetric.

Similarly,

$$C^t = \frac{1}{2}(A - A^t)^t = \frac{1}{2}(A^t - A) = -C,$$

so C is skew-symmetric

Therefore,

$$A = B + C \in W_1 + W_2,$$

which implies

$$V = W_1 + W_2.$$

□

proof of (2). Let

$$A \in W_1 \cap W_2.$$

Then

$$A^t = A, \quad A^t = -A.$$

Hence,

$$A = -A,$$

so

$$2A = \mathbf{0}.$$

Since the underlying field is $\mathbb{C}$,

$$A = \mathbf{0}.$$

Therefore,

$$W_1 \cap W_2 = \{\mathbf{0}\}.$$

□

Combining (1) and (2), we conclude that

$$V = W_1 \oplus W_2.$$

□